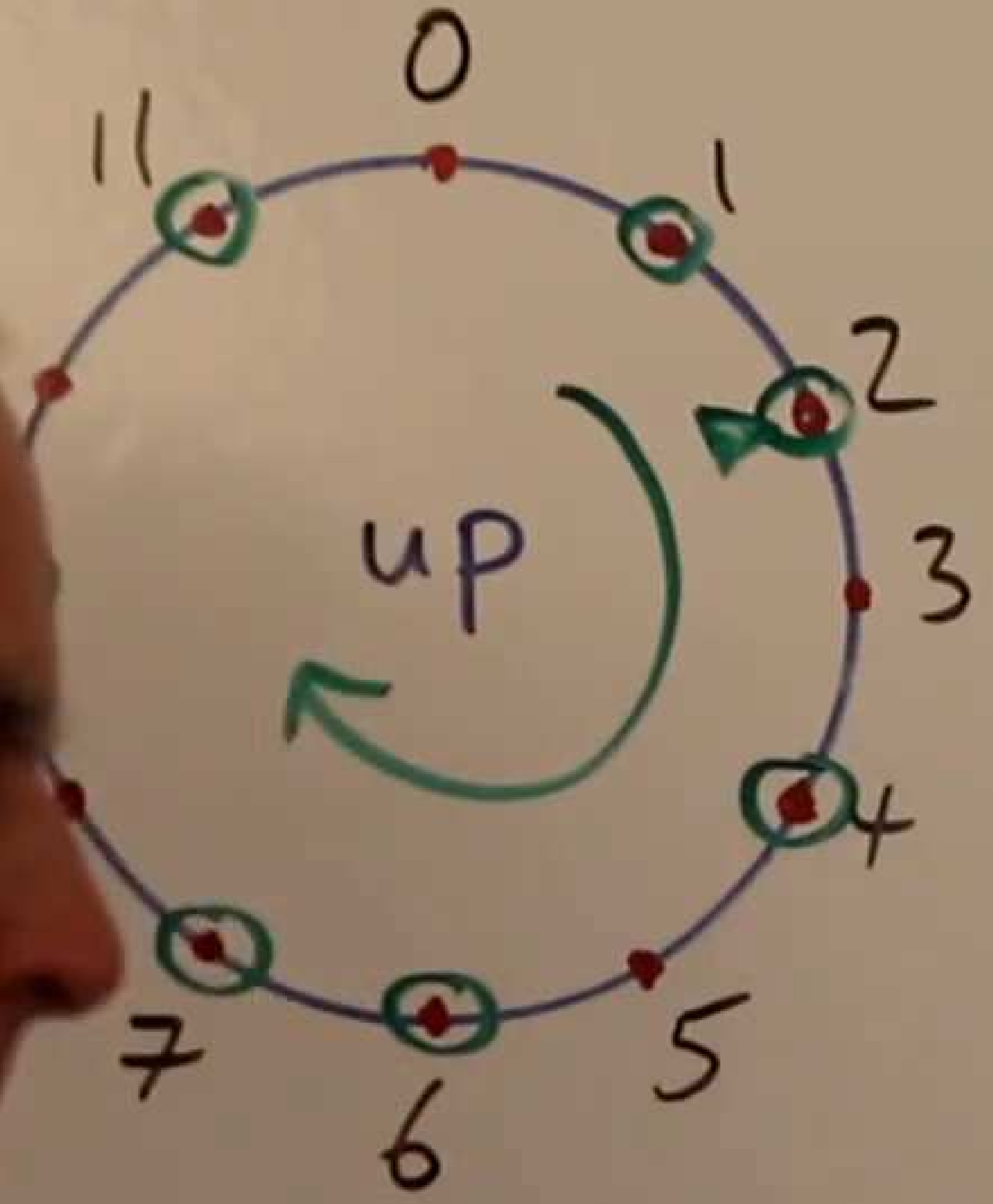
The mathematics of scales + modes
M1
M2
M3
M4
M8
The 12-tone chromatic scale (clock-scale)
M0, M1, ..., M11 + repeat
The interval between successive notes is a 1-step
①
le-types, modes
etry, reverse symmetry

Q. What is a scale? And how do we specify one?



Scale: An ascending / clockwise sequence of tones starting and ending on the same tone, & going around once.

Ex. $s = \langle 2, 4, 6, 7, 9, 11, 1, 2 \rangle$ ← pointy brackets

$\text{tonic}(s) = 2$ $|s| = 7$

③

Interval sequence of a scale:

records the successive differences

b lements of a scale

E <2, 4, 6, 7, 9, 11, 1, 2> then

2, 1, 2, 2, 2, 1] ← square brackets

N otal of elements

be 12 for any scale

0, 1, 3, 5, 6, 9, 10>

1, 2, 2, 1, 3, 1]

4


Scales s and t are of the same type
⟺ I(s) = I(t)
same type
0
11
1
10
2
9
3
4
8
7
6
5
⑤
<7,10,0,1,2,5,7>
I(t) = [3,2,1,1,3,2]
A scale like t = <7,10,0,1,2,5,7>
can also be specified by
I(t)=[3,2,1,1,3,2] ≡ I
tonic(t)
type are obtained
keeping the interval
and I(t') = I then
⑥

A scale like $t = \langle 7, 10, 0, 1, 2, 5, 7 \rangle$

also be specified by

c$(t) = 7$ and $I(t) = [3, 2, 1, 1, 3, 2] \equiv I$

s of the same type are obtained

ting the tonic, keeping the interval

nce. 6

If tonic$(t') = 4$ and $I(t') = I$ then

$t' = \langle 4,$

A scale like $t = \langle 7, 10, 0, 1, 2, 5, 7 \rangle$ can also be specified by

$\text{tonic}(t) = 7$ and $I(t) = [3, 2, 1, 1, 3, 2] \equiv I$

Scales of the same type are obtained by rotating the tonic, keeping the interval sequence.

6

Ex. If $\text{tonic}(t') = 4$ and $I(t') = I$ then

$$t' = \langle 4, 7, 9, 10, 11, 2, 4 \rangle$$

If we keep the tonic and rotate the interval sequence ... we get a mode!

Ex. ... , 0, 1, 2, 5, 7 >

⑦

toni... $Int(t) = [3, 2, 1, 1, 3, 2]$

The... create a mode of t by

tr... 7 $Int(\tilde{t}) = [2, 1, 1, 3, 2, 3]$